آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 8

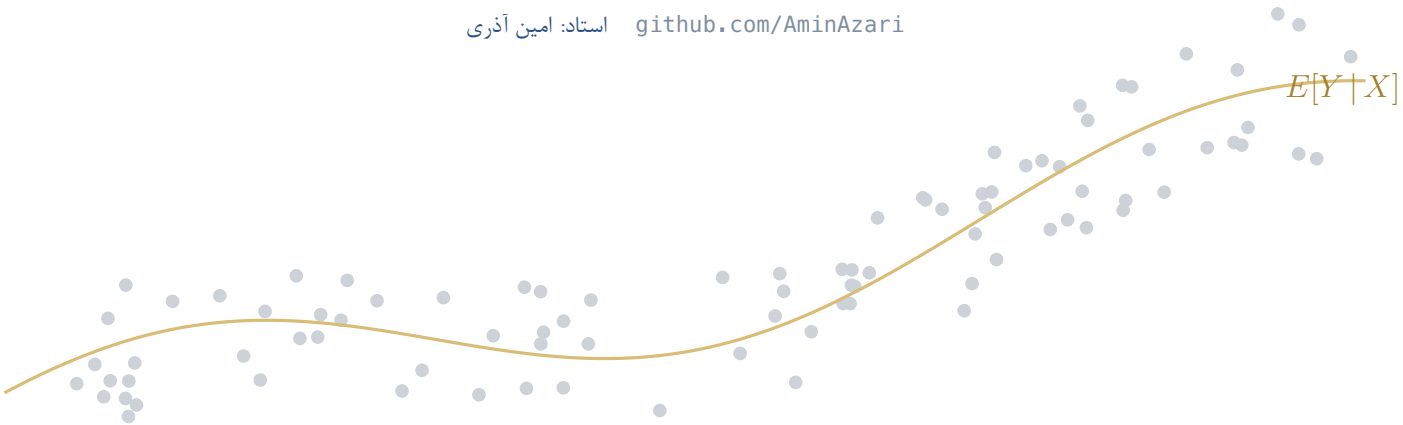
Conditional Expectation and Variance

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Learning Objectives

After completing this module, students will be able to:

1. Compute conditional expectations and variances
2. Apply the Law of Total Expectation
3. Apply the Law of Total Variance
4. Use these tools for engineering analysis across multiple operating conditions

8.1 Introduction

Many engineering systems operate under **different conditions** (channel states, modes, environments). We often need the **overall average performance** computed from conditional performance in each state.

- What is the average BER when the channel randomly switches between good and bad?
- What is the overall measurement variance when the sensor operates in different environments?

8.2 Conditional Expectation $E[X \mid Y = y]$

Definition

Continuous:

$$E[X|Y=y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y)\, dx$$

Discrete:

$$E[X|Y=y] = \sum_{x} x \cdot p_{X|Y}(x|y)$$

Interpretation

$E[X \mid Y = y]$ is the **average value of** X when we know $Y = y$. It is a function of y.

$E[X \mid Y]$ as a Random Variable

When Y is itself random, $E[X \mid Y]$ is a **random variable** — it takes the value $E[X \mid Y = y]$ when $Y = y$.

Engineering Example

Average received power given channel state:

- Good channel ($Y = 1$): $E[P \mid Y = 1] = 10$ mW
- Poor channel ($Y = 0$): $E[P \mid Y = 0] = 1$ mW
- $P(Y = 1) = 0.7$, $P(Y = 0) = 0.3$

8.3 Conditional Variance $\text{Var}(X \mid Y = y)$

Definition

$$\text{Var}(X|Y=y) = E[(X - E[X|Y=y])^2|Y=y] = E[X^2|Y=y] - (E[X|Y=y])^2$$

Interpretation

The conditional variance measures the **remaining uncertainty in** X after Y is known.

Engineering Example

Measurement precision depends on temperature:

- At $T = 20°C$: $\text{Var}(X \mid T = 20) = 0.01$ (high precision)
- At $T = 80°C$: $\text{Var}(X \mid T = 80) = 0.09$ (low precision due to thermal effects)

8.4 Law of Total Expectation (Tower Property)

Statement

$$E[X] = E[E[X|Y]]$$

Discrete version:

$$E[X] = \sum_y E[X|Y=y] \cdot P(Y=y)$$

Continuous version:

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y)\, dy$$

Proof (Discrete)

$E[E[X \mid Y]] = \sum_y E[X \mid Y=y] \cdot P(Y=y)$
$= \sum_y [\sum_x x \cdot P(X=x \mid Y=y)] \cdot P(Y=y)$
$= \sum_x \Sigma_y x \cdot P(X=x, Y=y)$
$= \sum_x x \cdot P(X=x) = E[X]$ ✔

Interpretation

The **overall average** equals the **weighted average of conditional averages**, weighted by the probability of each condition.

Engineering Example: Average BER Across Channel States

A wireless system switches between three modes:

- Mode 1 (60% of time): BER $= 10^{-6}$
- Mode 2 (30% of time): BER $= 10^{-4}$
- Mode 3 (10% of time): BER $= 10^{-2}$

Overall BER $= E[\text{BER}] = 0.6(10^{-6}) + 0.3(10^{-4}) + 0.1(10^{-2})$
$= 0.6 \times 10^{-6} + 30 \times 10^{-6} + 10000 \times 10^{-6} \approx 1.003 \times 10^{-3}$
Engineering insight: The overall BER is dominated by the worst mode, even though it occurs only 10% of the time.

Engineering Example: Average Lifetime with Defect Types

Components are defect-free (90%) or have a hidden defect (10%):

- Defect-free: $E[T \mid \text{good}] = 10000$ hours
- With defect: $E[T \mid \text{defect}] = 500$ hours

$E[T] = 0.9(10000) + 0.1(500) = 9050$ hours

8.5 Law of Total Variance

Statement

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

Term	Name	Interpretation
$\text{Var}(X)$	Total variance	Overall uncertainty in X
$E[\text{Var}(X \mid Y)]$	Expected conditional variance	Average "within-group" variability
$\text{Var}(E[X \mid Y])$	Variance of conditional means	"Between-group" variability

Proof

$\text{Var}(X) = E[X^2] - (E[X])^2$

$E[X^2] = E[E[X^2 \mid Y]]$ (tower property)
$= E[\text{Var}(X \mid Y) + (E[X \mid Y])^2]$
$= E[\text{Var}(X \mid Y)] + E[(E[X \mid Y])^2]$

$(E[X])^2 = (E[E[X \mid Y]])^2$

$\text{Var}(X) = E[\text{Var}(X \mid Y)] + E[(E[X \mid Y])^2] - (E[E[X \mid Y]])^2$
$= E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y])$ ✔

Interpretation: Decomposition of Uncertainty

Total uncertainty = Average uncertainty within each condition + Variability of the mean across conditions

Engineering Example: Sensor in Two Environments

A sensor operates indoors ($Y = 1$, prob 0.6) and outdoors ($Y = 2$, prob 0.4):

- Indoor: $E[X \mid Y = 1] = 25°C$, $\text{Var}(X \mid Y = 1) = 1$
- Outdoor: $E[X \mid Y = 2] = 15°C$, $\text{Var}(X \mid Y = 2) = 16$

**$E[\text{Var}(X \mid Y)] * * = 0.6(1) + 0.4(16) = 7.0$ (average within-condition variance)
**$\text{Var}(E[X \mid Y]) * * = E[(E[X \mid Y])^2] - (E[E[X \mid Y]])^2$
$= 0.6(25^2) + 0.4(15^2) - [0.6(25) + 0.4(15)]^2 = 375 + 90 - (21)^2 = 465 - 441 = 24$
**$\text{Var}(X) * * = 7.0 + 24.0 = 31.0$
Most variance comes from the **between-condition** difference (24 out of 31).

8.6 Practice Problems

Problem 1

A digital link operates over two channel types:

- Type A (prob 0.7): $E[\text{delay} \mid A] = 5\,\text{ms}$, $\text{Var}(\text{delay} \mid A) = 2$
- Type B (prob 0.3): $E[\text{delay} \mid B] = 20\,\text{ms}$, $\text{Var}(\text{delay} \mid B) = 25$

Find (a) $E[\text{delay}]$, (b) $\text{Var}(\text{delay})$.

Solution

(a) $E[D] = 0.7(5) + 0.3(20) = 3.5 + 6 = 9.5$ ms

(b) $E[\text{Var}(D \mid \text{type})] = 0.7(2) + 0.3(25) = 1.4 + 7.5 = 8.9$
$\text{Var}(E[D \mid \text{type}]) = 0.7(5^2) + 0.3(20^2) - 9.5^2 = 17.5 + 120 - 90.25 = 47.25$
$\text{Var}(D) = 8.9 + 47.25 = 56.15$

Problem 2

A random number N of components are tested. $N \sim \text{Poisson}(10)$. Each has independent failure probability $p = 0.05$. Let X = number of failures. Find $E[X]$ and $\text{Var}(X)$.

Solution $X \mid N \sim \text{Binomial}(N, 0.05)$. $E[X \mid N] = 0.05\,\text{N}$. $\text{Var}(X \mid N) = N(0.05)(0.95) = 0.0475\,\text{N}$.

$E[X] = E[E[X \mid N]] = E[0.05\,\text{N}] = 0.05(10) = 0.5$

$\text{Var}(X) = E[\text{Var}(X \mid N)] + \text{Var}(E[X \mid N]) = E[0.0475\,\text{N}] + \text{Var}(0.05\,\text{N}) = 0.0475(10) + 0.05^2(10) = 0.475 + 0.025 = 0.5$

Next Module: Module 9 — Covariance and Correlation